COURSE 2 · ALGEBRA

Algebra

Visual companion and Symbols Sheets

One section per module: the Symbols Sheet, with every numbered equation from the audio reader in ordinary notation, then the module's key diagram.

Keep it open while you listen, and beside you for the workbook.

Module 2.1 · Symbols and diagrams

Expressions and simplifying

2.1-1	Distributive law	$a(b+c) = ab + ac$
2.1-2	Expanding two brackets	$(a+b)(c+d) = ac + ad + bc + bd$
2.1-3	Index laws	$x^a x^b = x^{a+b}, \quad \frac{x^a}{x^b} = x^{a-b}, \quad (x^a)^b = x^{ab}$
2.1-4	Negative and fractional indices	$x^{-n} = \frac{1}{x^n}, \quad x^{1/n} = \sqrt[n]{x}$
2.1-5	Difference of two squares	$a^2 - b^2 = (a-b)(a+b)$

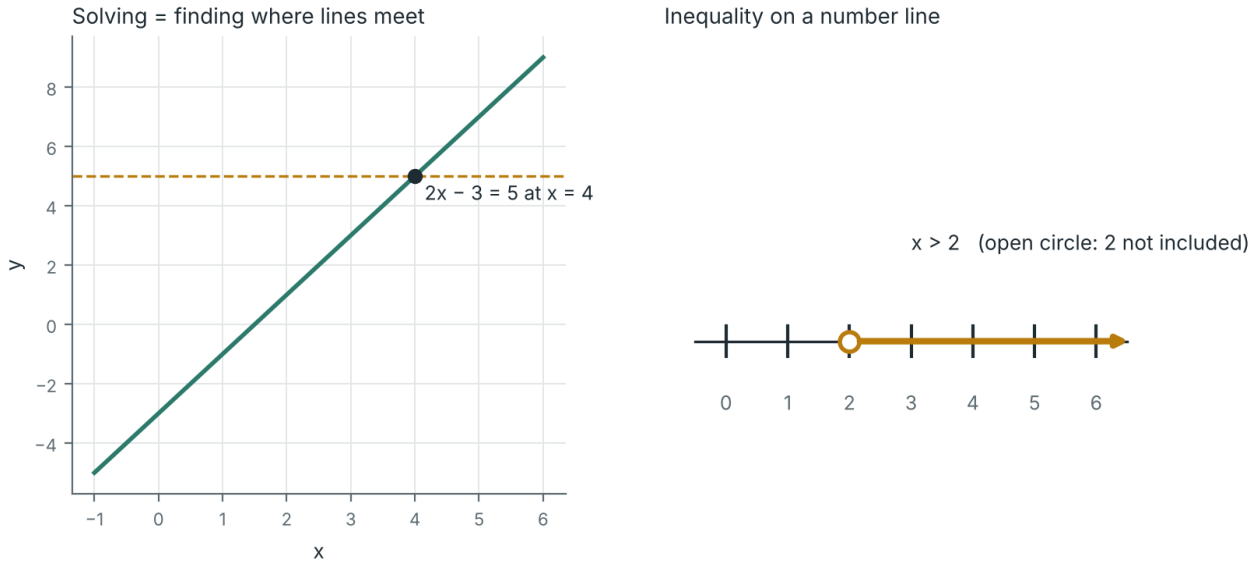
3(x + 4)	⇄	3x + 12	expand: multiply every term inside
x² + 5x + 6	⇄	(x + 2)(x + 3)	factorise: find two numbers, sum 5, product 6
2x + 3y − x + y	⇄	x + 4y	collect like terms

Expanding and factorising are two directions of the same identity.

Module 2.2 · Symbols and diagrams

Linear equations and inequalities

2.2-1	Straight line	$y = mx + c$
2.2-2	Inequality sign flip	$-x > 3 \;\Rightarrow\; x < -3$
2.2-3	Gradient between two points	$m = \frac{y_2 - y_1}{x_2 - x_1}$



Left: solving 2x − 3 = 5 is finding where y = 2x − 3 meets y = 5. Right: x > 2 shown on a number line.

Module 2.3 · Symbols and diagrams

Rearranging formulas

2.3-1	Simple pendulum	$T = 2\pi\sqrt{\frac{L}{g}}$
2.3-2	Rearranged for g	$g = \frac{4\pi^2 L}{T^2}$
2.3-3	Thin lens, rearranged for v	$\frac{1}{f} = \frac{1}{u} + \frac{1}{v} \Rightarrow v = \frac{uf}{u-f}$

$$T = 2\pi\sqrt{L/g} \xrightarrow{\div\, 2\pi} \frac{T}{2\pi} = \sqrt{\frac{L}{g}} \xrightarrow{\text{square}} \frac{T^2}{4\pi^2} = \frac{L}{g} \xrightarrow{\text{flip, } \times L} g = \frac{4\pi^2 L}{T^2}$$

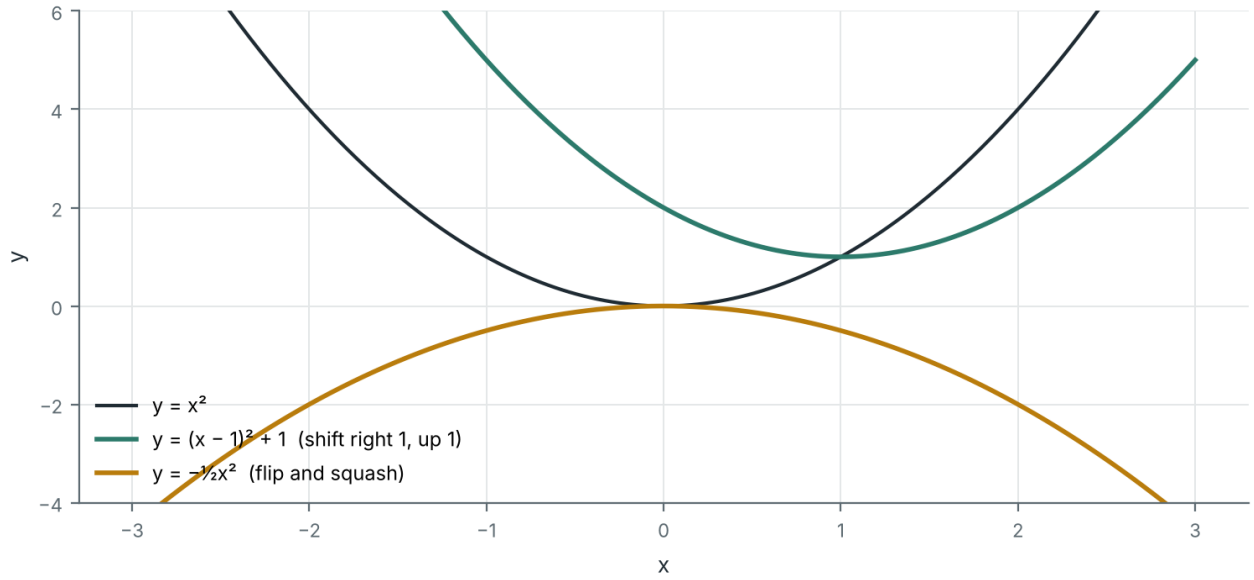
Outermost operation first: ×2π was applied last, so it is undone first. Each step acts on both whole sides.

Peeling the pendulum law for g: undo ×2π, then the root, then the division.

Module 2.4 · Symbols and diagrams

Functions and graphs

2.4-1	General transformation	$y = a\,f(b(x-h)) + k$
2.4-2	Vertex form of a parabola	$y = a(x-h)^2 + k$
2.4-3	Composite and inverse	$(f \circ g)(x) = f(g(x)), \qquad f^{-1}(f(x)) = x$

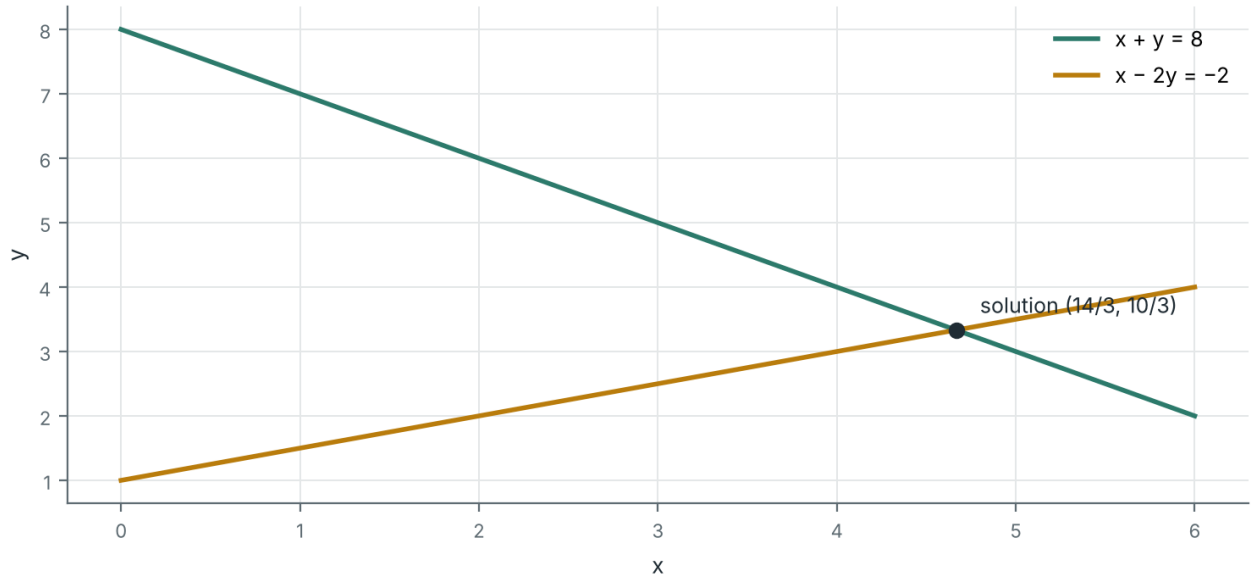


Transformations of $y = x^2$: shifting right and up, then flipping and squashing.

Module 2.5 · Symbols and diagrams

Systems of equations

2.5-1	General two-variable system	$a_1x + b_1y = c_1, \qquad a_2x + b_2y = c_2$
2.5-2	Elimination step	$(2x + 3y) - (2x - y) = 12 - 4 \;\Rightarrow\; 4y = 8$

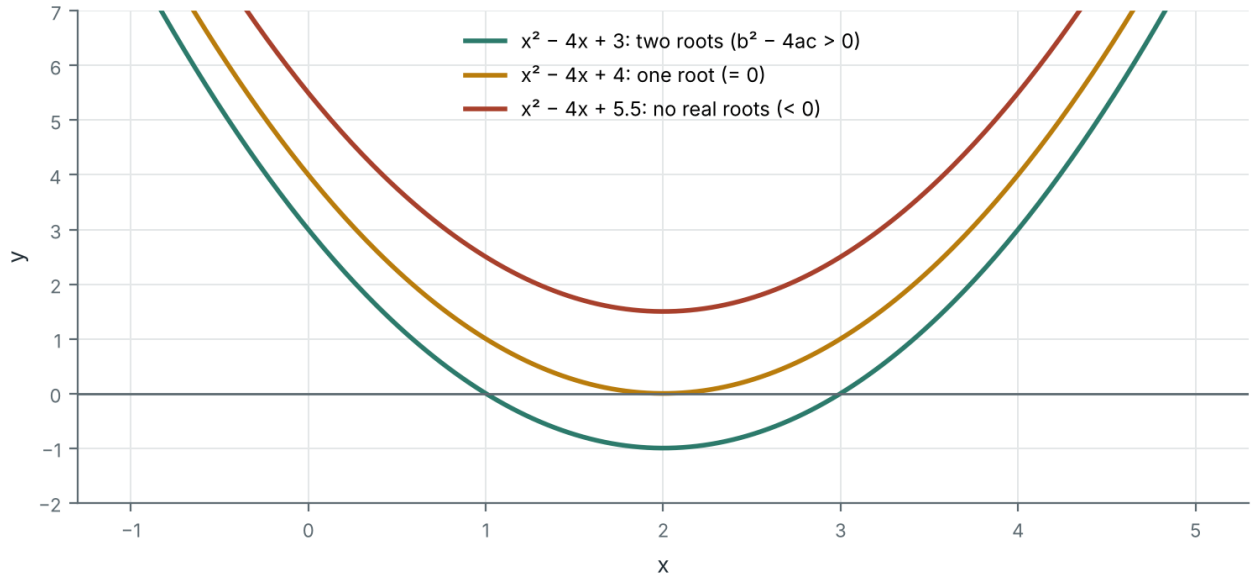


Two equations, two lines. The solution is where they cross.

Module 2.6 · Symbols and diagrams

Quadratics and polynomials

2.6-1	Quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
2.6-2	Discriminant	$\Delta = b^2 - 4ac$
2.6-3	Factor theorem	$p(k) = 0 \Leftrightarrow (x - k)$ is a factor
2.6-4	Vertex of $y = ax^2 + bx + c$	$x_{\text{vertex}} = -\frac{b}{2a}$

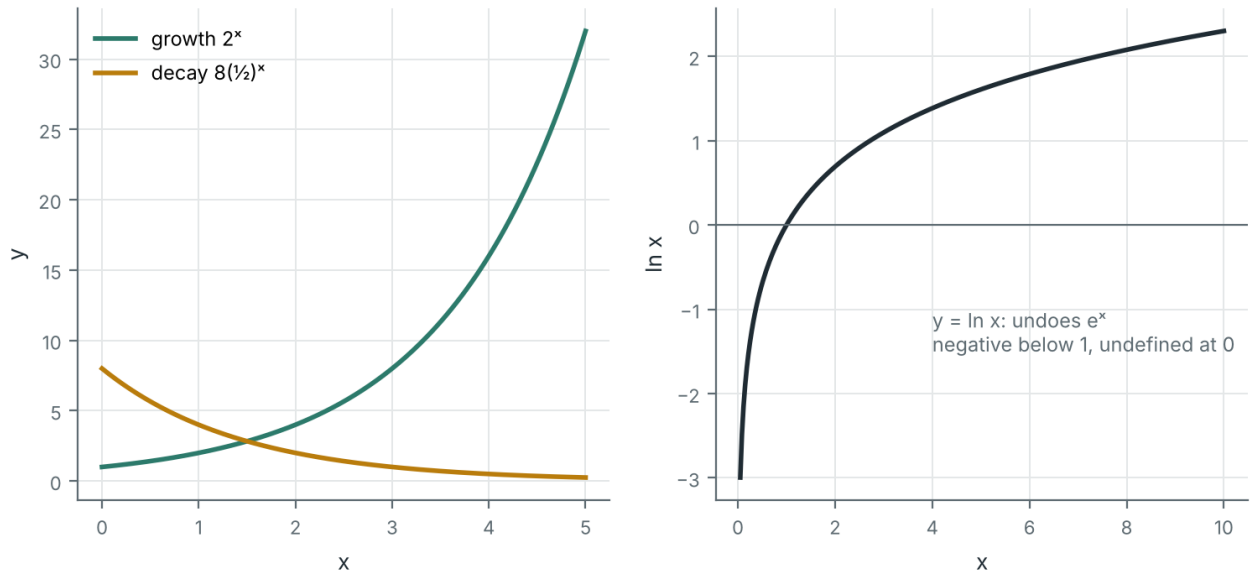


The discriminant decides how many times a parabola crosses the x-axis.

Module 2.7 · Symbols and diagrams

Exponentials and logarithms

2.7-1	Discrete growth	$N = N_0(1+r)^t$
2.7-2	Continuous growth and decay	$N = N_0 e^{kt}$
2.7-3	Log laws	$\log(ab) = \log a + \log b, \quad \log \frac{a}{b} = \log a - \log b, \quad \log a^n = n \log a$
2.7-4	Half-life	$t_{1/2} = \frac{\ln 2}{k}$
2.7-5	Solving an exponential	$a^x = b \Rightarrow x = \frac{\ln b}{\ln a}$

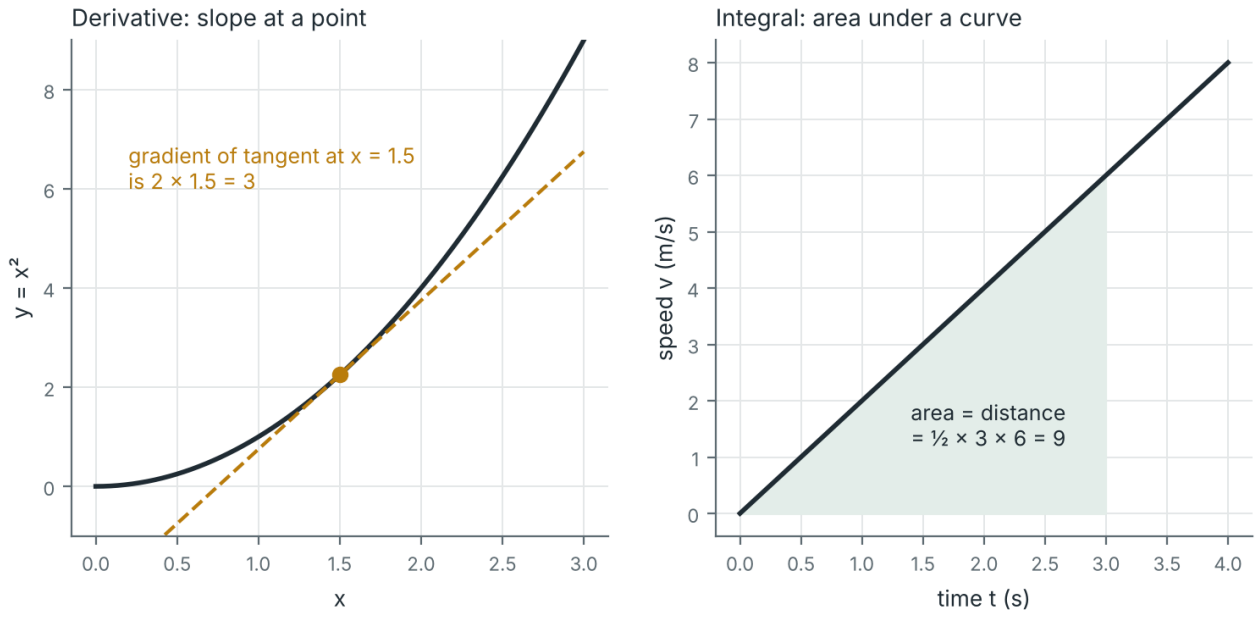


Left: exponential growth and decay. Right: the natural logarithm, the inverse of e^x.

Module 2.8 · Symbols and diagrams

Calculus intuition

2.8-1	Definition of the derivative	$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$
2.8-2	Power rule	$\frac{d}{dx} x^{n} = n x^{n-1}$
2.8-3	Two special derivatives	$\frac{d}{dx} e^{x} = e^{x}, \qquad \frac{d}{dx} \sin x = \cos x$
2.8-4	Area under a curve	$\int_{a}^{b} f(x)\, dx$
2.8-5	Integrating a power	$\int x^{n}\, dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$



Left: the derivative is the slope of the tangent. Right: the integral is the area under a curve; for a speed-time graph it is the distance travelled.